Impact of Floorplan Aspect Ratio on P&R Metrics in OpenLane

Utilization = 45 (constant for all 3 aspect ratios)

1. Aspect Ratio: 0.5

Simulation

```
sdudigani@sdudigani-VirtualBox:~/CDF_Tasks/OpenLane$ make mount
cd /home/sdudigani/CDF_Tasks/OpenLane && \
        docker run --rm -v /home/sdudigani/CDF_Tasks/OpenLane:/openlane -v /home/sdudigani/CDF_Tasks/OpenLane/designs:/openlane/install -v /home/sdudigani:/home/sdudigani -v /home/sdud
igani/.ciel:/home/sdudigani/.ciel -e PDK_ROOT=/home/sdudigani/.ciel -e PDK=sky130A  --user 1000:1000 -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix --network host --security-opt seccom
p=unconfined -v /home/sdudigani/.Xauthority:/.Xauthority -ti ghcr.io/the-openroad-project/openlane:ff5509f65b17bfa4068d5336495ab1718987ff69-amd64
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -interactive
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
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Available under the Apache License, version 2.0. See the LICENSE file for more details.

% package require openlane 1.0.1
1.0.2
% prep -design spm_ar05_clk2p5
[INFO]: Using configuration in 'designs/spm_ar05_clk2p5/config.json'...
[INFO]: PDK Root: /home/sdudigani/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/spm_ar05_clk2p5/runs/RUN_2026.06.09_20.20.30
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
```

Core Area

The following core area report has llx lly urx ury coordinates

```
5.52 10.88 133.86 73.44
/home/sdudigani/CDF_Tasks/OpenLane/designs/spm_ar05_clk2p5/runs/RUN_2026.06.09_20.20.30/reports/floorplan/3-initial_fp_core_area.rpt (END)
```

Routing Congestion

```
sdudigani@sdudigani-VirtualBox:~/CDF_Tasks/OpenLane/designs/spm_ar05_clk2p5/runs/RUN_2026.06.09_20.20.30$ grep -A15 "Final congestion report"  ~/CDF_Tasks/OpenLane/designs/spm_ar05_clk
2p5/runs/RUN_2026.06.09_20.20.30/logs/routing/18-global.log
[INFO GRT-0096] Final congestion report:
Layer         Resource        Demand        Usage (%)    Max H / Max V / Total Overflow
---------------------------------------------------------------------------------------
li1                  0             0            0.00%             0 /  0 /  0
met1              2648           372           14.05%             0 /  0 /  0
met2              2274           466           20.49%             0 /  0 /  0
met3              1703            19            1.12%             0 /  0 /  0
met4               800             9            1.12%             0 /  0 /  0
met5               190             0            0.00%             0 /  0 /  0
---------------------------------------------------------------------------------------
Total             7615           866           11.37%             0 /  0 /  0

[INFO GRT-0018] Total wirelength: 10764 um
[INFO GRT-0014] Routed nets: 381
[INFO GRT-0006] Repairing antennas, iteration 1.
[INFO GRT-0043] No OR_DEFAULT vias defined.
sdudigani@sdudigani-VirtualBox:~/CDF_Tasks/OpenLane/designs/spm_ar05_clk2p5/runs/RUN_2026.06.09_20.20.30$
```

Critical path slack

```
sdudigani@sdudigani-VirtualBox:~/CDF_Tasks/OpenLane/designs/spm_ar05_clk2p5/runs/RUN_2026.06.09_20.20.30$ grep -i "worst slack" ~/CDF_Tasks/OpenLane/designs/spm_ar05_clk2p5/runs/RUN_20
26.06.09_20.20.30/logs/signoff/30-rcx_sta.log
worst slack 0.81
worst slack 0.09
sdudigani@sdudigani-VirtualBox:~/CDF_Tasks/OpenLane/designs/spm_ar05_clk2p5/runs/RUN_2026.06.09_20.20.30$
```

2. Aspect Ratio: 1.0

Simulation

```
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -interactive
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
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Available under the Apache License, version 2.0. See the LICENSE file for more details.

% package require openlane 1.0.2
1.0.2
% prep -design spm_ar10_clk2p5
[INFO]: Using configuration in 'designs/spm_ar10_clk2p5/config.json'...
[INFO]: PDK Root: /home/sdudigani/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/spm_ar10_clk2p5/runs/RUN_2026.06.09_21.05.15
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
```

Core Area

```
5.52 10.88 96.14 100.64
/home/sdudigani/CDF_Tasks/OpenLane/designs/spm_ar10_clk2p5/runs/RUN_2026.06.09_21.05.15/reports/floorplan/3-initial_fp_core_area.rpt (END)
```

Routing Congestion

```
sdudigani@sdudigani-VirtualBox:~/CDF_Tasks/OpenLane/designs/spm_ar10_clk2p5/runs/RUN_2026.06.09_21.05.15$ grep -A15 "Final congestion report" ~/CDF_Tasks/OpenLane/designs/spm_ar10_clk2p5/runs/RUN_2026.06.09_21.05.15/logs/routing/18-global.log
[INFO GRT-0096] Final congestion report:
Layer         Resource        Demand        Usage (%)    Max H / Max V / Total Overflow
---------------------------------------------------------------------------------------
li1                  0             0            0.00%             0 /  0 /  0
met1              2266           312           13.77%             0 /  0 /  0
met2              2317           541           23.35%             0 /  0 /  0
met3              1495            15            1.00%             0 /  0 /  0
met4               793             6            0.76%             0 /  0 /  0
met5               182             0            0.00%             0 /  0 /  0
---------------------------------------------------------------------------------------
Total             7053           874           12.39%             0 /  0 /  0

[INFO GRT-0018] Total wirelength: 10639 um
[INFO GRT-0014] Routed nets: 381
[INFO GRT-0006] Repairing antennas, iteration 1.
[INFO GRT-0043] No OR_DEFAULT vias defined.
```

Critical path slack

```
sdudigani@sdudigani-VirtualBox:~/CDF_Tasks/OpenLane/designs/spm_ar10_clk2p5/runs/RUN_2026.06.09_21.05.15$ grep -i "worst slack" ~/CDF_Tasks/OpenLane/designs/spm_ar10_clk2p5/runs/RUN_2026.06.09_21.05.15/logs/signoff/30-rcx_sta.log
worst slack 0.82
worst slack 0.11
sdudigani@sdudigani-VirtualBox:~/CDF_Tasks/OpenLane/designs/spm_ar10_clk2p5/runs/RUN_2026.06.09_21.05.15$
```

3. Aspect Ratio: 2.0

Simulation

```
% prep -design spm_ar20_clk2p5
[INFO]: Using configuration in 'designs/spm_ar20_clk2p5/config.json'...
[INFO]: PDK Root: /home/sdudigani/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/spm_ar20_clk2p5/runs/RUN_2026.06.09_21.08.15
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
```

Core Area

```
5.52 10.88 69.46 138.72
/home/sdudigani/CDF_Tasks/OpenLane/designs/spm_ar20_clk2p5/runs/RUN_2026.06.09_21.08.15/reports/floorplan/33-initial_fp_core_area.rpt (END)
```

Routing Congestion

```
sdudigani@sdudigani-VirtualBox:~/CDF_Tasks/OpenLane/designs/spm_ar20_clk2p5/runs/RUN_2026.06.09_21.08.15$ grep -A15 "Final congestion report" ~/CDF_Tasks/OpenLane/designs/spm_ar20_clk2
p5/runs/RUN_2026.06.09_21.08.15/logs/routing/48-global.log
[INFO GRT-0096] Final congestion report:
Layer         Resource        Demand        Usage (%)    Max H / Max V / Total Overflow
---------------------------------------------------------------------------------------
li1                  0             0            0.00%             0 /  0 /  0
met1              2011           277           13.77%             0 /  0 /  0
met2              2320           712           30.69%             0 /  0 /  0
met3              1336             9            0.67%             0 /  0 /  0
met4               696            16            2.30%             0 /  0 /  0
met5               180             0            0.00%             0 /  0 /  0
---------------------------------------------------------------------------------------
Total             6543          1014           15.50%             0 /  0 /  0

[INFO GRT-0018] Total wirelength: 11978 um
[INFO GRT-0014] Routed nets: 384
[INFO GRT-0006] Repairing antennas, iteration 1.
[INFO GRT-0043] No OR_DEFAULT vias defined.
```

Critical path slack

```
sdudigani@sdudigani-VirtualBox:~/CDF_Tasks/OpenLane/designs/spm_ar20_clk2p5/runs/RUN_2026.06.09_21.08.15$ grep -i "worst slack" ~/CDF_Tasks/OpenLane/designs/spm_ar20_clk2p5/runs/RUN_20
26.06.09_21.08.15/logs/signoff/60-rcx_sta.log
worst slack 0.70
worst slack 0.11
sdudigani@sdudigani-VirtualBox:~/CDF_Tasks/OpenLane/designs/spm_ar20_clk2p5/runs/RUN_2026.06.09_21.08.15$
```

Summary of key metrics

Final Core Area Analysis

	AR=0.5	AR=1.0	AR=2.0
core width (μm)	128.34	90.62	63.94
core height (μm)	62.56	89.76	127.84
core area (μm^2)	8024.15	8134.05	8170.09

Note: core width = urx- llx, core height = ury-lly, core area = width x height

- Changing the aspect ratio significantly altered the geometry of the floorplan.
- As the aspect ratio increased from 0.5 to 2.0, the design changed from a wide layout to a tall layout. The core width decreased from 128.34 μm to 63.94 μm, while the core height increased from 62.56 μm to 127.84 μm.
- Although the utilization remained constant, a slight increase in total core area was observed which indicates that floorplan shape can influence the final dimensions and area allocation of the design.

Routing Congestion Analysis

	AR=0.5	AR=1.0	AR=2.0
resource	7615	7053	6543
demand	866	874	1014
usage	11.37%	12.39%	15.50%

- Routing congestion increased as the aspect ratio increased.
- The AR = 2.0 floorplan exhibited the highest routing demand and congestion because the taller layout required longer vertical interconnect paths and concentrated routing resources within a narrower width.

- The AR = 0.5 configuration showed the lowest routing utilization, indicating a more relaxed routing environment.
- No routing overflow was observed in any configuration, demonstrating that all floorplans were routable and met the routing requirements successfully.

Critical path slack Analysis

	AR=0.5	AR=1.0	AR=2.0
worst setup slack	0.81	0.82	0.70
worst hold slack	0.09	0.11	0.11

- Critical path slack was analyzed using post-route signoff timing reports.
- The square floorplan (AR = 1.0) achieved the best timing margin with the highest setup slack of 0.82 ns.
- The AR = 2.0 configuration showed the lowest setup slack (0.70 ns), indicating that increased routing demand and longer interconnect paths can negatively impact timing performance.
- All three designs satisfied timing requirements since both setup and hold slacks remained positive.

Observations

- Increasing aspect ratio reduced floorplan width and increased floorplan height.
- Routing demand and congestion increased with larger aspect ratios.
- The square floorplan (AR = 1.0) provided the best balance between routing efficiency and timing performance.
- The tall floorplan (AR = 2.0) exhibited the highest routing utilization and the lowest setup timing margin.

References

[1] OpenLane Documentation – https://openlane.readthedocs.io

[2] OpenROAD Project – https://theopenroadproject.org

[3] SKY130 PDK Documentation – https://skywater-pdk.readthedocs.io